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A Framework of Effective Science Explanation Videos Informed by Criteria for Instructional Explanations

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Abstract

Science explanation videos, especially online ones, have become popular. They cover nearly all topics from school science curricula. Teachers use them in a flipped classroom. Students watch them also in their leisure time. However, there is a lack of research on guidelines that effective science explanation videos should follow. The research on instructional explanations provides essential insights into the effectiveness of science explanation videos. The goal of the present article is to develop a framework for effective science explanation videos. The development consists of two steps. First, criteria for a high explaining quality are derived from the research on effective instructional explanations. That leads to a coherent theoretical framework that requires empirical justification. The second step, therefore, is an exploration of the effects of the framework. We developed two explanation videos: one high explaining quality (HE) video following the framework and one low explaining quality (LE) video explicitly not following the framework. Both videos, however, are scientifically correct and provide the same learning opportunities. Students from a German high school assigned to two groups ($n = 90$; $n = 86$) watched, respectively, the HE and LE videos, and participated in tests for declarative and conceptual knowledge before and after watching the videos. The main result is that the HE group outperformed the LE group in the achievement of declarative knowledge (Cohen's $d = .42$, $p = .007$). We interpret that as an argument for the validity of the framework of effective science explanation videos presented in this article.

Keywords Explanation videos · Instructional videos · Instructional explanations · Achievement · Explanatory videos · Flipped classroom

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Introduction

Explanation videos have become increasingly popular, especially because they are easily accessible via online platforms like YouTube. Nearly all topics mentioned in school science curricula are covered by videos on YouTube, most of which have multiple versions. In fact, science is one of the most frequent domains in explanation videos of all (Wolf and Kratzer 2015). Furthermore, surveys suggested that students of all ages use these online videos frequently for various purposes, for example, for completing their homework or preparing for exams (Wolf 2018). Also, for formal learning, explanation videos are used; in some flipped classroom settings, for example, when preparing at home for a topic, students use videos and work with the information later to solve problems in classroom (Bishop and Verleger 2013; Schmidt and Ralph 2016; Thomson et al. 2014). Explanation videos have become a challenge for current science education research concerning questions of how to implement such videos in science teaching and how to develop explanation videos on science topics (Schmidt-McCormack et al. 2017). Wolf and Kulgemeyer (2016) suggest that explanation videos can contribute to science education in various ways (Table 1). Wolf (2018) refers to Table 1 and argues that tasks, in which students produce their own videos, have high educational value. It challenges the students' content knowledge and fosters communication skills to develop a video that explains a given topic to (younger) peers. Some teachers develop their own explanation videos for flipped classroom settings. Wolf (2018) also acknowledges that the analysis of such videos can be useful in science teaching, both for teachers (e.g. to learn from other expert explainers) and for students (e.g. to see alternative explaining approaches that complement their teachers' explanations). Wolf (2018) highlights that, in particular, *online* explanation videos are a welcome addition because teachers and other professional explainers such as publishers of teaching material no longer have a monopoly on providing instructional explanations.

We regard it as likely that the importance of explanation videos for science teaching will increase over the coming years. For all the possibilities to use science explanation videos shown in Table 1, a framework is required that helps to analyse the quality of science explanation videos and helps teachers to select high-quality videos for their students. Such a framework should also give teachers, students and other producers of videos hints on how to develop science explanation videos. However, it lacks research on criteria for explanation

Table 1 Examples of the potential use of explanation videos in science classroom (as suggested by Wolf and Kulgemeyer 2016)

	Producer of explanation video	
Explainee	Teachers	Students
Teachers	Teachers learn from other explaining experts how to explain a particular topic (e.g. good examples)	Teachers use students' explanation videos as an alternative form of assessment
Students	Students learn from expert explainers, an additional explaining approach compared to their own teacher's is accessible Teachers develop explanation videos for their own students and use them in flipped classroom settings	Students produce explanation video; teachers assist them by giving feedback on (a) scientific correctness and (b) appropriateness of video for a particular group of interest. (Learning opportunity for (a) content knowledge and (b) communication skills)

videos. Currently, published research reports are mostly about best practice examples (Stockwell et al. 2015) and short review studies on how students use explanation videos (Brame 2016). Also, research on digital media in science teaching provides hints on how to develop these videos (Beheshti et al. 2018). The "cognitive theory of multimedia learning" (Mayer 2001) and the "cognitive model of multimedia learning" by Schnotz et al. (cf. Schnotz and Bannert 2003) have been used to design several kinds of learning environments, including videos. However, to date, there has been little agreement on whether or not the frameworks from multimedia learning can be used without any adaptation to developing explanation videos—some scholars see a need for criteria beyond these frameworks (Schroeder and Traxler 2017).

As far as the authors know, no specific framework for the production of (online) science explanation videos has been developed *and* empirically tested yet. Most likely, the development and the analysis of most science explanation videos on YouTube follow everyday concepts on teaching and learning and not evidence-based and theoretically consistent criteria. Therefore, there is a need for an evidence-based framework that provides information on how to develop science explanation videos both for teachers and for producers of explanation videos in general. This framework should also provide criteria to analyse the quality of explanation videos. Also, there is a need for empirical research to gain more profound insights into how explanation videos work. Interestingly, in science education, very few studies have investigated the primary communicative situation of explanation videos: an instructional explanation that has been filmed. In our point of view, the question of how to develop science explanation videos is at the core of science education research—after all, the primary purpose of science explanation videos is providing clear and comprehensible instructional explanation of a scientific topic.

Most surprisingly, there is a lack of connection between the research on the quality of instructional explanations and the research on explanation videos, especially those on online videos (Kulgemeyer and Peters 2016). We see high potential in such a connection because the research on effective instructional explanations might provide the criteria that are essential to produce and analyse explanation videos.

Our study, therefore, has the goal to develop a framework of effective science explanation videos. The development consist of two steps:

(1) The theoretical development of a coherent framework of effective online explanation videos based on criteria for effective instructional explanations. The framework should provide guidelines on how to develop online explanation videos.
(2) An empirical exploration of the effects of this framework on student achievement. Based on the analysis of the literature in step 1, we made assumptions about the differential effects of science explanation videos that have been developed according to the framework. In the second step, we investigate the effects of the framework and collect arguments for the validity of the framework based on these assumptions.

In the first part of the article that reports the theoretical part of our study, we discuss which of the criteria for effective instructional explanations that have been described in empirical studies over the last years might apply to online explanation videos. We derive the theoretical framework of the explaining quality of explanation videos based on this discussion. In the second part that reports the empirical part of our study, we use this framework in an explorative study and compare the effects of two videos on student achievement. The first video was produced following the criteria of the framework. The second video, which is also

scientifically correct, provided identical learning opportunities for students who used it for the same duration and its text had the same readability; however, its production did not follow the criteria of effective explaining as described in the framework.

Theoretical Background

Explanation Videos

Explanation videos are short videos (usually between 5 and 10 min) with the goal to explain a particular content area (Wolf and Kratzer 2015). In recent research, they have alternatively been referred to, for example, as educational videos (Brame 2016), instructional videos (Schroeder and Traxler 2017) or explanatory videos (Kulgemeyer and Peters 2016). Even though there has been research on factors for the increased popularity of these videos, most of the studies on the design principles for explanation videos have been conducted in the context of multimedia learning (Mayer 2014). In a general analysis of the literature since 1957, Pekdag and Le Marechal (2010) point out the potential of videos for chemistry education, including the creation of a productive learning environment. They also highlight that videos (or more specific: explanation videos) can be used to prepare for exams and even to replace written explanations in textbooks (Pekdag and Le Marechal 2010). However, it needs high-quality explanation videos to contribute to teaching.

Brame (2016) described guidelines on how to optimise learning from videos based on cognitive load theory and active learning. Also, Muller (2008) used criteria from multimedia learning to study explanation videos. In fact, cognitive load theory probably provides the most frequently used criteria to describe effective digital learning environments, including videos. Initially described by Sweller (1988), cognitive load theory has become very influential for multimedia learning in general. The cognitive theory of multimedia learning (Mayer 2001) is based on cognitive load theory. The basic idea is that new information has to pass through working memory to be stored in long-term memory. Working memory, however, has limited capacities. For any content, three kinds of *cognitive load* might use these capacities. The *intrinsic load* is part of the content itself and relies on the connections of the elements that belong to the content. The *germane cognitive load* is determined by the degree of cognitive processes required to reach a goal. There are cognitive processes that require more cognitive capacities than others. The third kind of cognitive load is the *extraneous load*. Extraneous load is not helpful for learning intention and might result from distracting information or a confusing learning environment. For learning purposes, it would be helpful to focus the mental capacities on the content. DeJong (2010) discussed recent studies on cognitive load that have implications for the development of learning environments, particularly those that require multimedia. The cognitive theory of multimedia learning acknowledges that there are two channels for information, both of which have independent but limited capacities (*dual coding theory*, Paivio (1986)). Again, these channels can be described using cognitive load theory (Mayer and Moreno 2003). Information might pass via the visual channel or the auditory channel. Brame (2016) derived criteria for effective videos from this theoretical framework and proposed, for example, ways to reduce the extraneous cognitive load by eliminating unnecessary information (*weeding*) or to enhance the germane load by using both auditory and visual channels.

However, debate continues about the validity of cognitive load theory. De Jong (2010) highlights that the main construct of the theory—the cognitive load—is not well-defined, especially with respect to its three components. For example, the extraneous cognitive load and the germane cognitive load are hard to distinguish: "Methods that prima facie seem to induce extraneous load may also induce germane load and vice versa" (Moreno 2010, p. 136). Also, to date, there is no reliable and valid measure of the cognitive load in general (Moreno 2010). Besides, questions have been raised about whether or not cognitive load theory can be applied to real-life situations—which would question that the theory can be used to design instruction (Sorden 2013).

Besides Brame's (2016) framework and Muller (2008), the question of how to develop effective online science explanation videos on platforms like YouTube has not been researched extensively. There were just a few studies on the impact of online explanation videos on student achievement, but their results are promising (cf. Zhang et al. 2006). Brame (2016) provided a good overview of what the research on digital media had proposed to produce such videos. We firmly believe that the focus on instructional explanations adds a valuable perspective for producing effective science explanation videos—even more since it is unclear whether or not the frameworks from multimedia learning apply to explanation videos (Schroeder and Traxler 2017).

Throughout this article, the term *explanation videos* refers to short videos of approximately 6 min as suggested by Brame (2016) with the aim to make an audience understand a principle. That is very similar to those videos found on platforms like YouTube (Kulgemeyer & Peters 2016). Thus, explanation videos in this article do not comprise tutorials or instructions on how to solve a particular type of problems (e.g. a demonstration of how to use Lagrangian mechanics) (Wolf and Kratzer 2015). The reason for this definition is provided by the concept 'instructional explanation' which is described in the following section.

Instructional Explanations

Instructional explanations—which are sometimes also referred to as teaching explanations or teacher explanations—are communicative attempts to *engender comprehension* (Gage 1968). In prior studies, instructional explanations have been distinguished from both scientific explanations and a transmissive view of teaching and learning.

Scientific explanations are often reduced to causal explanations connecting a phenomenon logically with an underlying principle, which predicts the phenomenon's occurrence regarding antecedent conditions (Hempel and Oppenheim 1948). However, other possible scientific explanations have been discussed, for example, statistical explanations (Rescher 1962), teleological explanations (Talanquer 2007) and the idea of improving scientific explanations as a framework for teaching and learning (Papadouris et al. 2017). Science education research has also been studying scientific explanations for many years, often regarding their structure compared to argumentation (cf. Berland and McNeill 2012; Berland and Reiser 2008). However, scientific explanations and instructional explanations are not identical (Treagust and Harrison 1999). The most important difference is their intention. Instructional explanations aim to make science content comprehensible for an explainee, whereas this addressee of the communication is not of primary interest for a scientific explanation. It has been shown that an instructional explanation likely to be understood has to consider the explainee's needs, whereas an explanation solely guided by the content structure is less effective (Nathan and Petrosino 2003). It is a common misunderstanding that an instructional explanation equals a scientific

explanation presented in a clear language (Kulgemeyer & Schecker 2009; Wittwer and Renkl 2008). In fact, this transmissive view on the explaining process underestimates the possible potential of instructional explanations—explaining is a process that requires interaction between explainer and explainee, and most importantly, adaptation to the explainee's needs (Kulgemeyer & Riese 2018; Wittwer and Renkl 2008). These kinds of instructional explanations potentially contribute to high-quality science teaching (Geelan 2012). This interactive way of explaining requires a diagnosis of understanding (Kulgemeyer & Schecker 2013) and is a highly challenging practice (Charalambous et al. 2011). Actually, even very well-crafted explanations can never ensure that an explainee understands the science content. They increase the likelihood that an explainee constructs meaning from the explained information (Kulgemeyer & Schecker 2013).

What Makes Instructional Explanations Effective?

The question of what makes instructional explanations effective has been discussed in previous empirical studies, most prominently from a psychological perspective in Wittwer and Renkl's (2008) review. In science education, Geelan (2012) published a review article on essential studies on instructional explanations in science teaching. Kulgemeyer and Schecker (2013) researched high school students explaining skills based on a model of dialogic explaining. Sevian and Gonsalves (2008) proposed a framework that describes scientists' explanations to the public. These studies identified some general factors that have contributed to the effectiveness of an instructional explanation. In the following, we describe these factors based on empirical evidence and discuss their applicability to explanation videos.

Many studies have researched the structure of an explanation regarding possible impacts on its effectiveness. Seidel et al. (2013), for example, compared the effectiveness of two possible approaches: (1) mentioning the principle at first and illustrating it with appropriate examples later (rule-example strategy) and (2) showing an example at first and deriving the principle from it afterwards (example-rule strategy). Seidel et al. also reported on differential effects of these two approaches. The rule-example strategy seems to be the better choice if the learning intention of the explanation is factual knowledge, whereas the example-rule strategy is superior if the goal is learning a routine of how to solve unique problems. Korthagen and Kessels (1999) and Champagne et al. (1982) came to similar results. When these approaches are applied to science teaching, it would mean that learning a scientific principle from an explanation would likely be more successful if the explanations follow the rule-example strategy. Learning how to solve a unique problem such as calculations in a particular area like energy should be more successful if the explanations follow the example-rule strategy. These two strategies—inductive versus deductive explaining—are certainly possible factors that influence the effectiveness of explanation videos.

Another feature of the structure of an explanation that has been discussed regarding its effectiveness is the occurrence of summaries (Kulgemeyer & Schecker 2013). According to research, summaries, in general, might contribute to the effectiveness of an explanation. However, they should not be given too frequently. An important feature of successful explanations is for them to avoid digressions while keeping the focus on the core ideas (*minimal explanations*) (Anderson et al. 1995; Renkl et al. 2006). The reason for this effect has often been explained by using the idea of cognitive load (cf. Mayer et al. 1999; Sweller and Chandler 1991). Redundancies use cognitive resources, which are no longer available for

working with the core ideas. That also applies to the use of summaries. For novices, summaries contribute to a better understanding, whereas for experts, summaries are not required and they contribute more to an unnecessary cognitive load (Kalyuga 2007).

Anderson et al. (1995) considered effective explanations as minimal, while *minimal* means such explanations are precise, free from irrelevant details and coherent. Indeed, coherence is another factor that has been discussed in the context of effective explanations. Coherence means that an explanation assists an explainee in constructing a mental representation of the explained content by providing mental connections (Kamalski et al. 2008). An indicator of a high coherence would be the connection of two separate sentences by one single noun appearing in both of them (a lexical tie). Also, conjunctions help to connect sentences, for explanations, casual connectives such as *because* in particular (Lehman and Schraw 2002). For example, Wittwer and Ihme (2014) showed the importance of coherence for understanding written explanations. However, we consider that coherence is likely also important for verbal explanations and explanation videos. Coherent videos should, for example, avoid naming the same object differently, even though experts might consider the different names to be synonyms. Knowing about synonyms requires knowledge that novices do not have, therefore, this results in *inferences*. Of particular importance is the connection between principles and examples with the causal connective *because* as it helps to identify the underlying principle.

Most of the literature in educational psychology dealing with the effectiveness of explanations probably regards *adaptation to knowledge prerequisites* as the most important factor (Wittwer and Renkl 2008). Duffy et al. (1986) found that outstanding teachers use their diagnosis as a basis for further adaptation of explanations. Wittwer et al. (2010) showed that computer experts were able to improve the quality of their explanations by using given information for adaptation. For science teaching, adaptation is crucial as well, and even more so because science education research has dealt with the identification of frequent misconceptions for several decades. The model of dialogic explanation in science, therefore, considers adaptation to prior knowledge, to misconceptions, but also to the explainee's interests as vital (Kulgemeyer & Schecker 2009; Kulgemeyer & SheckerAuthors 2013). This model further elaborates what adaptation means by providing four tools, which can be used to adapt an explanation to an explainee's prerequisites. First, the *language level* (with a range from everyday language to scientific language) might be varied. Second, the level of *mathematization* (e.g. from qualitative descriptions to formulas) should be adapted. Third, appropriate *representation forms and demonstrations* beyond verbal language (e.g., from photos to animation, diagrams and beyond to items and even experimental setups) should be used. Last, the use of *examples*, *models* and *analogies* plays an essential role and should bridge the gap between new information and familiar content by connecting these two areas. Studies show that these four tools are used for adaptation by both students explaining as peer tutors (Kulgemeyer & Schecker 2013) and teachers (Kulgemeyer & Tomczyszyn 2015).

Whereas the importance of adaptation for instructional explanations is widely accepted, it is probably the feature most challenging to regard in the production of explanation videos. Instructional explanations in science teaching are considered to be a process that requires the provision of an initial explanation, diagnosis and further adaptation that might lead to a better-adapted explanation. This process can only occur in face-to-face communication. Wolf and Kulgemeyer (2016) discussed this as a core problem for the potential effectiveness of explanation videos. It leads to an important restriction in the use of such videos. Explanation videos might be produced for a particular group of interest, but their approach cannot be changed to better meet the group's needs during the process of being used in an explanation.

They might, therefore, fail in their effective use for many explainees. The four tools for adaptation can be used in explanation videos, but there is no process of diagnosis and further refinement of the explanation. Wittwer and Renkl (2008) highlighted "the importance of a precise mental model of the individual learner for designing effective instructional explanations" (Wittwer and Renkl 2008, p. 52). It is even more important for the production of an explanation video to contain reliable information about prior knowledge, interests and possible misconceptions of the group of explainees. During an instructional explanation, the explainer's mental model of the explainee's characteristics improves, however, the video is necessarily adapted to the explainee's characteristics defined before its production. The explainee, however, might search for alternative videos, which might lead to adaptation that is guided by the explainee. It is a challenge for future research to investigate the competencies necessary to choose a video that best meets an individual's needs.

Another feature of successful explanations that has been researched is the idea of highlighting the relevant parts, which is important for two reasons. First, the perceived relevance of an explanation seems to be crucial to motivating students to engage in deeper processing of the information. Acuña et al. (2011) showed that this affects the effectiveness of remedial explanations, which are explanations that focus on mistakes, for example. Examining the effects of highlighting the relevancy of an explanation, Sánchez et al. (2009) found that learners who received the information that an explanation deals with a common misconception outperformed those who just received a rough explanation. A promising way to highlight the relevancy are prompts that point to the most important parts and the essential ideas of an explanation (Chi et al. 2001; Roelle et al. 2014).

Next to adaptation, the most critical feature of instructional explanations is their integration into cognitive activities (Wittwer and Renkl 2008). Instructional explanations should not replace knowledge construction activities. Altmann and Nückles (2017) reported that problem-solving activities affect the effectiveness of an explanation. Learning tasks that require the explained information for solving a scientific problem seem to be a promising feature. Even more, some studies showed that these learning tasks are more important than the instructional explanation itself (Webb et al. 2006). Renkl et al. (2006) stressed that even well-designed instructional explanations frequently fail if the explained information is not required for follow-up learning tasks. That means that after receiving an explanation, learners are probably able to repeat the content of an explanation, but they are not necessarily able to use this knowledge for further problem-solving. In a nutshell, they have declarative knowledge, but they cannot transfer it to other context areas. Renkl et al. (2006) argued that understanding a scientific principle includes a cognitive scheme that enables an individual to solve different problems using the same principle. For example, a scientific principle like Newton's third law applies to many different situations. After being given an instructional explanation, learners might be able to repeat the law and to show how it is used in the examples that have been named in the explanation (declarative knowledge). Without follow-up learning activities, they, however, do most likely not learn how to apply the concept to alternative situations (they need conceptual knowledge to do so). We regard this idea to be essential for explanation videos. For effective use of explanation videos, these videos should end with a learning task. Alternatively, science teachers should develop these learning tasks when they use the videos. The results from the research on instructional explanations support the idea of the use of videos in flipped classroom settings.

For the empirical part of this article, we use the idea that instructional explanations without learning tasks only affect the explainees' declarative knowledge. An important prerequisite for an effective instructional explanation is that the explanation focuses on concepts or principles

(Renkl et al. 2006). Alty and Coombs (1981) showed that explainers sometimes tend to explain step-by-step how to solve a particular problem instead of pointing out the underlying principle. Dutke and Reimer (2000) found that including the principle explicitly is more effective than explaining a simple step-by-step solution. That also supports the idea that highlighting the most relevant parts of an explanation is essential for its effectiveness. For explanation videos, that would mean that the videos should explicitly focus on a principle. For example, they should focus on Newton's third law instead of just showing step-by-step how the law can be used to calculate the acceleration of a rocket.

A Framework of Effective Explanation Videos

Considering the results of the studies discussed above, we developed a framework for effective explanation videos. The current state of the literature on instructional explanations suggests seven factors that influence the effectiveness of instructional videos: (1) the structure of the video, (2) the adaptation to a group of addressees, (3) the use of appropriate tools to achieve adaptation, (4) minimal explanations that avoid digressions and keep the cognitive load low, (5) highlighting that the explanation itself is relevant to the learner and what the most relevant parts are, (6) providing learning tasks subsequent to the video and (7) a focus on a scientific principle. These seven factors can be achieved using different features. Table 2 presents an overview of the framework. In this study, we refer to criteria for effective instructional explanations and their application to developing explanation videos. We acknowledge that the research on multimedia learning also contributes to the effectiveness of explanation videos as suggested in prior research (e.g., Brame 2016; Muller 2008). An obvious connection between our framework and the research on multimedia learning is the importance of cognitive load theory common to both. However, the focus of this study lies in the application of criteria of effective instructional explanations. A detailed discussion of the differences and commonalities between this field of research and the research on multimedia learning is beyond the scope of this article.

Research Questions and Hypotheses

The empirical part of this article aims to test the theoretical framework empirically. It addresses the following research question:

> *Research question:* Does an explanation video developed by following a high fit to the framework lead to higher student achievement compared to an explanation video developed by following a low fit to the framework?
>
> *Hypothesis 1:* For both the high explaining quality (HE) and the low explaining quality (LE) video groups, a significant gain in their declarative knowledge can be revealed.
> *Hypothesis 2:* The HE group outperforms the LE group regarding their gain in declarative knowledge, but not their gain in conceptual knowledge.

We understand Hypothesis 1 as a prerequisite for the whole study. The most challenging part of this study was to develop two videos that can provide identical learning opportunities and are both scientifically correct but still differ in their fit to the framework

Table 2 Overview of the framework for effective explanation videos

Factors	Feature	Description
Structure	1. Rule-example, example-rule	If the learning goal is factual knowledge, the video follows the rule-example structure If the learning goal is a routine or procedural knowledge, the video follows the example-rule structure
	2. Summarising	The video summarises the explanation
Adaptation	3. Adaptation to prior knowledge, misconceptions and interest	The video adapts the explanation to a well-described group of addresses and their potential knowledge, misconceptions or interests. To do so, it uses the 'tools for adaptation'
Tools for adaptation	4. Examples	The video uses examples to illustrate a principle
	5. Analogies and models	The video uses analogies and models that connect the new information with a familiar area
	6. Representation forms and demonstrations	The video uses representation forms or demonstrations
	7. Level of language	The video uses a familiar level of language
	8. Level of mathematization	The video uses a familiar level of mathematization
Minimal explanation	9. Avoiding digressions	The video focuses on the core idea, avoids digressions and keeps the cognitive load low. In particular, it avoids using too many 'tools for adaptation' or summaries
	10. High coherence	The video connects sentences with connectors, especially 'because'
Highlighting relevancy	11. Highlighting relevancy	The video highlights explicitly why the explained topic is relevant to the explainee
	12. Direct addressing	The explainee is getting addressed directly, e.g. by using the second-person singular instead of the passive voice
Follow-up learning tasks	13. Follow-up learning tasks	The video describes learning tasks the explainees can engage with to actively use the new information after the video
New, complex principles	14. New, complex principle	The video focuses on a new science principle that is too complex to understand by self-explaining, e.g., because there are frequent misconceptions

described above. The effect of the framework was the one that we wanted to research. Thus, we wanted to control other possible confounding variables that might cause a higher knowledge gain. We, therefore, developed the videos in such a way that they both use language, terms, representation forms and mathematics the participants of the study could understand in general (according to the curricula of this part of Germany), but we acknowledge that not all confounding variables could be controlled. However, we used Hypothesis 1 as a quantitative hint that the videos for both groups were developed in such a way that students could potentially learn the whole content with each video. If the videos are comparable in their learning opportunities, a valid test instrument should diagnose learning gains for both of them. That is, from our point of view, an important prerequisite for our goal to identify the possible effects of the developed framework on student learning gains.

Conceptual knowledge means, in this case, the knowledge that can be transferred from the context in the explanation video to another context. According to the theory of instructional explanations, it was unlikely that the participants in this study could gain

conceptual knowledge without the following phase in which they applied their knowledge self-guided to solving other problems (learning tasks). We, therefore, assumed that no significant difference between the HE and LE groups could be found. Declarative knowledge in this study was considered as the knowledge that is closely connected to the context of the videos. Here, we expected a difference. This declarative knowledge is the knowledge students might use in the follow-up learning tasks for developing their conceptual knowledge.

Methods

Design of the Study

We developed two explanation videos that differ in their explaining quality based on the framework described above. They, however, are both scientifically correct, their scripts have comparable readability and their duration is nearly the same (for details, see the section below). They also both provide opportunities for learning the knowledge covered by the items of the test instrument.

We used a lesson of 45-min duration for the study. We formed two comparable experimental groups and administered to them a test instrument before and after they watched either one of the two videos. The test instrument included both items for conceptual and declarative knowledge. For each group, the 45 min included 30 min for the tests (15 min each for the pre- and post-test), approximately 7 min for watching the videos and 8 min for the instructions about research participation. The two experimental groups only differed in one way, that is, the video provided for them to watch.

Participants

The participants of this study were students ($n = 176$) from eight classes of a high school in northern Germany. Four of the classes were from 9th grade and four from 10th grade. We randomly assigned each two intact classes from the 9th grade and two from the 10th grade to one of the two experimental groups (i.e. HE and LE groups). We were aware that this approach resulted in a clustered sampling. That, indeed, was a limitation of the study, but it was impossible to assign single students randomly to one of the two experimental conditions.

The high explaining quality video or HE group consisting of four classes ($n = 90$) and the low explaining quality video or LE group also consisting of four classes ($n = 86$) did not differ significantly in their prior knowledge for both declarative knowledge and conceptual knowledge (for all t tests, $p > .05$). They were also comparable regarding their age and gender. Table 3 gives an overview of the characteristics the two groups shared.

Development of the Videos for the HE and LE Groups

For this study, it is of great importance that the explanation videos only differ in whether or not their production was following the framework described above so that important confounding variables are controlled. For example, both videos should be scientifically correct, should provide the same time on task and the same opportunities for learning the knowledge covered by the test items.

Table 3 Comparison of the high-explaining-quality video or HE group and the low-explaining-quality (LE) group concerning their demographics and pre-test measures

Dependent variables	Range	HE group Mean (SD)	LE group Mean (SD)	Cohen's *d**
Age		15.01	15.03	− .02
Gender	0 = male 1 = female	.53	.56	− .06
Class	9–10	9.44	9.51	− .14
Declarative knowledge (pre)	0–100%	8.0	7.9	.05
Conceptual knowledge (pre)	0–100%	46.5	50.8	− .18

*All reported differences are nonsignificant ($p > .05$)

Topic, Context and Basic Structure of the Explanation Videos

The topic of the explanation for both videos was the same and in German. For both videos, the explained topic was the phenomenon that cars frequently slide around a sharp curve after heavy rain that caused water built-up on the surface of the road. We decided to choose this rather complex topic because we had much experience with video studies on teachers' face-to-face explanations to students regarding this topic (Kulgemeyer & Riese 2018). We had, therefore, already observed a broad range of various approaches to explain this topic and felt confident that this topic allows variations in the quality of explaining while still being scientifically correct. Besides, for this topic, we were able to produce explanation videos with the quality close to that of authentic instructional explanations which is often the case for online explanation videos (Kulgemeyer and Peters 2016). To explain this phenomenon, three physics concepts are required and mentioned in both videos in the same order: inertia, the mechanics of circular motion and friction. The basic structure of both videos was as follows:

(1) Both videos start with a brief description of the problem. Both, therefore, follow an example-rule structure.
(2) Inertia is identified as the reason why a force is needed to drive a car around a curve.
(3) The curve is identified as part of a circle and, therefore, the force needed is a centripetal force.
(4) The centripetal force is the frictional force between tyres and the surface of the road.
(5) Finally, a difference in the coefficient of friction between the material pairs—asphalt and water tyres—explains why the cars slide on a particular spot. The limit of the centripetal force is lower with water on the road. However, the lower the centripetal force is, the higher the possible curve radius will be.

We assumed the explainees to be average physics students from grade 9 of German high schools, that is, they were interested in everyday contexts and knew about inertia and the concept of force as the reason for acceleration in general. They, however, never heard about friction or centripetal force before.

We developed two scripts before the production of the video: one for a video with a high fit to the framework of effective explanation videos (video 1, high explaining quality) and one for a video with a low fit to the framework (video 2, low explaining quality). As the basic structure of the videos was identical, we were able to vary how the explanation video was produced

following the framework. We were, however, also aware that video 2 (low explaining quality) should fulfil some criteria of effective explanation videos for a fair comparison.

In both videos, we filmed examples of a teacher using representation forms, a whiteboard and, in the case of video 1, short demonstrations to illustrate the topic. Many explanation videos on YouTube were also produced in that way (Kulgemeyer and Peters 2016). We, therefore, kept the videos close to many examples that can be found online with the goal that we could also research our framework for its applicability to other online explanation videos.

Shared Elements and Unique Features of the Explanation Videos

First, to ensure the comparability of the videos, we developed scripts for the instructional explanations that included text, representation forms and possible demonstrations. We then discussed the scripts in a group of two physics educators and identified the opportunities for learning the knowledge needed to solve the test items. That was the first approach to ensure that the same learning opportunities were provided in both videos.

Second, we also wanted to ensure that both videos are comparable in their general complexity of language. The use of scientific language is part of the framework (tools for adaptation), but the general complexity (e.g. unnecessarily long and complex sentences) would be an important confounding variable. We, therefore, calculated readability scores of the texts. We used the frequently used formulas Flesch Reading Ease Index (FRE) (Flesch 1948) and Kincaid Grade level (KG) (Kincaid et al. 1975) for this purpose. FRE is a measure that is based on the average sentence length and the average number of syllables per word. We used Flesch's formula in a version adapted to the German language. KG is based on the same measure but results in a score that reflects the grade in which students would be supposed to understand a text. Even though understanding a text is a process that involves both knowledges of the reader and the text, readability scores give first hints of the text complexity (Crossley et al. 2017). Both scripts were found to have comparable readability scores (for video 1, FRE = 67 indicating an easy text in the German version, and KG = 10 indicating that students from the tenth grade should understand the text; for video 2, FRE = 6 and KG = 10).

Third, we recorded the explanation videos based on the scripts. We ensured that both videos are comparable in their duration (video 1: 6:31 min; video 2: 6:46 min). In a group of eight educators familiar with criteria of explaining quality, we discussed the videos to ensure that their fit to the framework was their only different feature and reached a consensus on that. Table 4 shows an overview of the shared features and the unique features of the two videos.

We want to highlight the main differences between the videos. Video 1 (HE) has a better adaptation to the assumed prior knowledge regarding the *tools for adaptation*. It also has a better fit to the factor minimal explanation due to its higher coherence, in particular, because it connects the used examples by highlighting the similarities. Also, video 1 has better use of prompts that highlight the most relevant parts of the explanation and manages to address the explainees directly by using the second person singular instead of passive voice (factor *highlighting the relevancy*). Last but not least, video 1 includes a summary at the end of the video (factor *structure*).

The shared features include identical learning opportunities, the duration and the readability. Also, both videos concern new and complex principles (factor *new, complex principles*) because the topic was identical. Both videos also share an example-rule structure because, due to the shared topic, they need to start from the description of the problem (factor structure).

Table 4 Comparison of the high-explaining-quality (HE) video or video 1 and the low-explaining-quality (LE) video or video 2 concerning their fit to the framework (criteria 1 to 13), readability of their scripts and duration. Italic = unique features. Normal font = commonalities

	Feature	Video 1 (HE)	Video 2 (LE)
1	Rule-example	Example-rule	Example-rule
2	*Summarising*	*One at the end*	*No summary*
3	*Adaptation to prior knowledge, misconceptions and interest*	*Adapted to assumed knowledge of grade 9. Higher adaptation concerning the tools for adaptation (criteria 4–8)*	*Adapted to assumed knowledge of grade 9. Lower adaptation concerning the tools for adaptation (criteria 4–8)*
4	Examples	Yes	Yes
5	*Analogies and models*	*Yes*	*No*
6	*Representation forms and demonstrations*	*Yes (*e.g. *whiteboard, sketch of the curve, force diagrams, photos)*	*No (just the whiteboard and a sketch of the curve)*
7	*Level of language*	*Describing scientific terms in everyday language*	*Giving scientific definitions for scientific terms (according to the textbook these particular students use in physics)*
8	*Level of mathematization*	*Using qualitative paraphrase for mathematical relationship: 'the more …, the more …'*	*Using formulas (from the textbook these students use in physics)*
9	Avoiding digressions	Yes	Yes
10	*High coherence*	*Showing similarities between examples*	*No particular focus on coherence*
11	*Highlighting relevancy*	*Pointing out relevancies by using prompts*	*No*
12	*Direct addressing*	*Using second person singular (and imperative)*	*Using passive voice*
13	Follow-up learning tasks	No	No
14	New, complex principle	Yes	Yes
	Duration	6:31 min	6:46 min
	Readability of the texts	Flesch-Reading-Ease (German): 67 (easy)	Flesch-Reading-Ease (German): 66 (easy)
		Kincaid grade level = 10	Kincaid grade level = 10

Both videos avoid digressions (factor minimal explanations) and do not include a follow-up learning task due to a need for effective use of testing time.

In a nutshell, video 1 has a higher fit to the framework than video 2 (Table 4). We, however, acknowledge that our study cannot reveal information about which of the features contributes the most to an effective video. All of the unique features might contribute to a possible effect. We regard this as appropriate for a first exploration.

Instruments

We developed a short test (15 min) to examine our hypotheses (for sample items, see Table 5). The test consisted of seven items covering the conceptual knowledge (multiple choice) of the involved concepts (inertia, centripetal force, friction) and an open item for the declarative knowledge of the explanation. The three concepts important to understanding the phenomenon were covered by the items, which are a representative sample for the content validity of the scale (inertia, 1 item; centripetal force, 3 items; friction, 3 items). As we did not want to differentiate between the concepts, we treated the seven items as a single scale. The reliability was found to be low ($\alpha = .52$). However, the average inter-item correlation ($x = .16$) was found

just to fall within the range recommended by Clark and Watson (1995) for appropriate internal consistency ($.15 < x < .5$). We argue that a scale comprising three very different concepts with a broad range of misconceptions will never reach high reliability. We, therefore, regard the scale as appropriate for a first exploration of the framework, which was the second goal of our study. A higher number of items would have required more time. In this case, we would have had problems to find enough participants because the time would have exceeded one lesson in school. We, however, acknowledge the low reliability as a major limitation of our study. We discuss consequences that may result from this limitation at the end of this article.

Declarative knowledge was tested with a single open item that asked for the repetition of the explanation from the video. We coded the answers of the students by awarding points. One point was awarded for mentioning a concept of importance in understanding the phenomenon (one of the three as described above: *inertia*, *centripetal force*, *friction between surfaces*). An additional point for each concept was awarded if the concept was not only mentioned but also supported by a reason why it is important in this context. For example, "inertia is the reason why a car travels with a constant velocity" would be awarded one point. "Inertia is the reason why a car travels with a constant velocity. In this case, it changes its direction at the corner. Therefore, a force acts on the car." would be awarded two points, one for the concept and one for the reason. Both the examples were students' real answers. Forty of the 176 answers were coded by two raters, who agreed in all the cases in which points were awarded for particular answers.

Findings

Hypothesis 1: Significant Gain of Declarative Knowledge for both Treatment Groups

First, we analysed the test results of the high explaining quality (HE) video group and the low explaining quality (LE) video group for the differences between their scores in the pre- and post-test measures concerning both declarative and conceptual knowledge. This analysis has been conducted in two steps: first, to compare the gains in declarative knowledge and second, to compare the gains in conceptual knowledge. Because we want to research the knowledge gains for each group separately, for both steps, we used each two paired *t* test comparisons of the scores in the pre- and post-test, one for the HE group and one for the LE group. Using paired *t* test, we compared for each the HE and LE group the average of their scores (in percent) in the pre- and post-test. In the following, we report the average score in percent (*M*)

Table 5 Sample items from the test instrument (conceptual knowledge and declarative knowledge, translated from German)

Sample item 'conceptual knowledge'
Sheldon plays with his model train. Unfortunately, he drives the train too fast around a curve to the left. What happens to the train?
(a) The train derails and falls to the left
(b) The train derails and goes straight
(c) The train derails and goes around a curve to the right
(d) It does not matter how fast the train goes; it cannot derail as long as the rails are not broken
Item 'declarative knowledge'
Give a brief explanation in your own words why a car is more likely to slide around a curve if water is on the road than if the road is dry!

and the standard deviation SD of this score as well as the values of the *t* test. As a measure of the effect size, we use Cohen's *d*.

The first analysis of the two treatment groups concerning the declarative knowledge gains revealed a significant difference between the pre- and the post-test measures for both the HE and LE groups. For the HE group, the results of the paired *t* test comparison were pre-test, $M = 8.0\%$, $SD = 15.8\%$; post-test, $M = 28.9\%$, $SD = 28.0\%$; $t(178) = 6.2$, $p < .001$, Cohen's $d = .92$. Therefore, the students in the HE group increased their test scores significantly from 8 to 28.9%. For the LE group, the results of the paired *t* test comparison were pre-test, $M = 7.9\%$, $SD = 16.0\%$; post-test, $M = 18.1\%$, $SD = 23.7\%$; $t(174) = 2.74$, $p < .01$, Cohen's $d = .42$. Therefore, the students in the LE group increased their test scores significantly from 7.9 to 18.1%.

The same *t* test analysis was conducted on conceptual knowledge test results, revealing a significant difference between the pre- and the post-test measure for both the HE and LE groups. For the HE group, the results were pre-test, $M = 46.5\%$, $SD = 22.7\%$; post-test, $M = 70.7\%$, $SD = 22.0\%$; $t(170) = 3.32$, $p < .01$, Cohen's $d = .50$. Therefore, the students in the HE group increased their test scores significantly from 46.5 to 70.7%. For the LE group, the results were pre-test, $M = 50.8$, $SD = 22.8\%$; post-test, $M = 69.4\%$, $SD = 17.1\%$, $t(170) = 6.05$, $p < .001$, Cohen's $d = .92$. Therefore, the students in the HE group increased their test scores significantly from 50.8 to 69.4%. To sum up, both treatment groups showed significant gains of knowledge regarding both their declarative and conceptual knowledge.

Hypothesis 2: HE Group Outperforms LE Group Regarding their Gain of Declarative Knowledge Gain but Not their Gain of Conceptual Knowledge

To analyse this hypothesis, we first compared the two treatment groups regarding the post-test measures. We used independent *t* test for this purpose. The analysis revealed that the HE group (post-test declarative knowledge: $M = 28.9\%$, $SD = 28.0\%$) outperformed the LE group (post-test declarative knowledge $M = 18.1\%$, $SD = 23.7\%$) ($t(174) = 2.74$, $p < .01$, Cohen's $d = .42$). Therefore, the students in the HE group had a significantly higher declarative knowledge in the post-test compared to the students in the LE group. We, however, did not find a significant difference regarding the post-test measures in their conceptual knowledge (see Table 6).

For further analysis, we included the pre-test measures to understand whether or not the post-test scores reflect the gain of knowledge appropriately. Both groups should not differ in their pre-test scores. Figure 1 shows the results of the analysis. According to the *t* test, both treatment groups did not differ significantly regarding the pre-test measures of declarative and conceptual knowledge. However, regarding the post-test measure of declarative knowledge, we identified a significant difference between the two groups. The HE group outperformed the LE group (Cohen's $d = .42$, $p = .007$). Based on the benchmarks suggested by Cohen (1988), the effect size reached a medium effect size ($.3 < d < 4$) and based on Hattie's (2009)

Table 6 Comparison of the HE and LE groups concerning post-test measures

Dependent variables	Range	HE group *M* (SD)	LE group *M* (SD)	Cohen's *d*
Declarative knowledge (post)	0–100%	28.9 (28.0)	18.1 (23,7)	.42*
Conceptual knowledge (post)	0–100%	70.7 (22.0)	69.4 (17.1)	.06

*$p < .01$

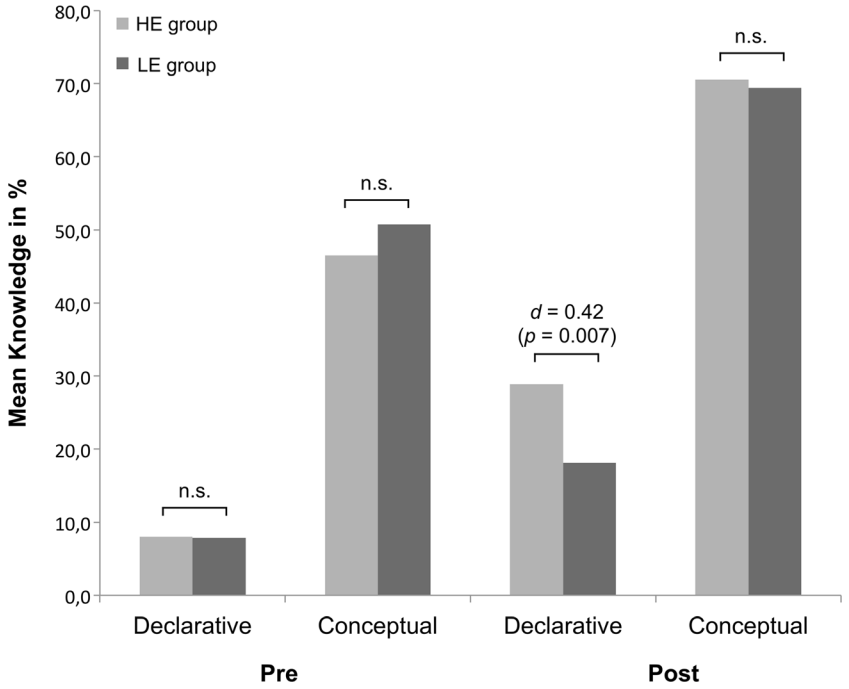


Fig. 1 Comparison of declarative and conceptual knowledge of the HE and LE groups measured by their pre- and post-test regarding *t* test significance differences with effect size

benchmark, it reached the zone of desired effects ($d > .4$). The post-test measures on conceptual knowledge did not differ significantly between both the two treatment groups.

Discussion

The goal of this study was to develop a framework for effective science online explanation videos. The *first step* to achieve this goal was the theoretical development of the first framework for effective online explanation videos based on the research on instructional explanations. We developed a framework that consists of 14 criteria (see Table 2) that seem promising as guidelines for the development of online explanation videos.

The *second step* of this development was a first empirical exploration of the framework. This exploration should provide arguments for the validity of the framework. We explored possible differential effects of the framework for effective explanation videos on students' declarative and conceptual knowledge. In this step, declarative knowledge means non-transferable knowledge that is limited to the content of the explanation video. This kind of knowledge is assumed to develop into conceptual knowledge potentially. Conceptual knowledge is transferable knowledge of scientific concepts that enables an individual to solve different problems using the relevant scientific principles.

The current state of research on instructional explanations suggests that it is difficult to achieve students' gains in conceptual knowledge without a learning task after an instructional explanation—even though the empirical basis for this assumption is not entirely convincing.

Still, we assumed that the framework of effective explanation videos does not affect students' achievement regarding their gains in conceptual knowledge. It should, however, affect their gains in declarative knowledge.

The most challenging part of the study was the development of the videos. We had to ensure that other meaningful variables did not confound the effects of the framework. The shared elements of both videos (HE/video 1 and LE/video 2) included the duration and readability scores of their texts. It took us much time and effort to develop the two videos in such a way that it is not obvious that one of them provides better learning opportunities. We, however, acknowledge that possible quality differences beyond the reported variables were an important limitation of our study. Our study should be considered as an exploration of the field because there had not been any prior studies that applied quality criteria of instructional explanations to the development of explanation videos in science. One indicator that both explanation videos provide comparable learning opportunities for the concepts tested with the instruments was analysed by testing Hypothesis 1. Students from both treatment groups learned significantly from the explanation videos regarding the gains in their declarative knowledge. That was why we accepted Hypothesis 1. Accepting Hypothesis 1 was the prerequisite for further analysis. The students from both groups, however, also gained significantly in their conceptual knowledge from the videos. Based on prior studies, we did not expect such results. In particular, two possible reasons for this phenomenon may apply: (1) the low reliability of the scale caused the effect, which was not present or (2) the students were reminded by the videos of concepts that they had the already known but were not part of their active knowledge. The second possibility does not sound very convincing because the science curriculum of this part in Germany covers the concepts of interest later. The students most likely did not yet have learning opportunities for them.

In accordance with the current state of the literature, we still believe that short explanation videos just slightly influence students' conceptual knowledge, particularly because many students probably hold misconceptions about centripetal force.

For both conceptual and declarative knowledge, it is uncertain whether or not these knowledge gains of the student would persist, as we did not conduct a follow-up measurement. Based on prior studies, we, however, believe that if we had wanted to investigate persistent knowledge gains, a follow-up task would have been needed in this study. We suggest that further research on this aspect should be conducted.

Regarding Hypothesis 2, we found the expected results and accepted the hypothesis. The treatment group that watched video 1 (high explaining quality) significantly outperformed the treatment group that watched video 2 (low explaining quality) regarding the post-test results on their declarative knowledge (Cohen's $d = .42$, $p = .007$). However, as we expected, we could not find a significant difference between the two groups in the post-test results on their conceptual knowledge.

Another limitation of the study was its design being a quasi-experimental study. Even though the classes were assigned randomly to a treatment group, the individuals were still assigned based on their classes as intact groups. We, however, regard this approach as appropriate for a first exploration of the effects of the framework.

It is important to highlight for further studies that we decided to vary many variables of the framework between the videos to test the framework as a whole but not on a single isolated factor. For a first exploration of the framework, we argue that this is an appropriate approach. However, future studies might provide insights into which of the factors that could account for

the differences between the two videos. We still do not know which of the factors that would contribute to the main effect.

We conclude that the present article presents a coherent theoretical framework that has been tested empirically. The empirical results, in general, support the framework, which makes it a promising starting point in an emerging field of research. We can show that explanation videos following the guidelines of the framework affect the development of descriptive knowledge in general. Of course, implementing explanation videos into effective science teaching means more than just watching them. There is a need to develop learning tasks that help to integrate explanation videos into ongoing cognitive activities (cf. Wittwer and Renkl 2008). According to the theory, learning tasks should be a key to develop conceptual knowledge by using explanation videos as well.

What do our results mean for the development of explanation videos? We believe that science education researchers are just beginning to understand the criteria for high-quality explanation videos, particularly those online ones. We argue that our results showed that the research on effective instructional explanations provides a promising framework for explanation videos. We, however, also acknowledge that other fields of research, most prominently the research on multimedia learning and the research on the popularity of online videos, probably also contributes to developing a more useful framework. It is an important goal of further research in science education to find commonalities and differences between these approaches with the goal to develop a joint framework. We strongly believe that during the coming years, digital media in general and online explanation videos, in particular, will become increasingly important for science education both in formal and informal learning. Our framework might serve as a starting point to develop high-quality online explanation videos for science teaching. Also, we argue that student learning in flipped classroom settings might benefit from our framework.

To conclude, we agree with Wolf and Kulgemeyer (2016) and Nielsen and Hoban (2015) that it is important for teachers to have students watch explanation videos—but maybe it is even more efficient for the students to develop their own videos. Also, in this respect, we regard our framework as a promising starting point that might help to produce these video in such a way that they are not only scientifically correct but also focus on the explainee's needs. However, we are aware that further research is needed and our framework even might be outdated soon in an area of science education, which is as rapidly developing as digital media.

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